

Optimizing surfaces to enhance cryogenic boiling for space applications

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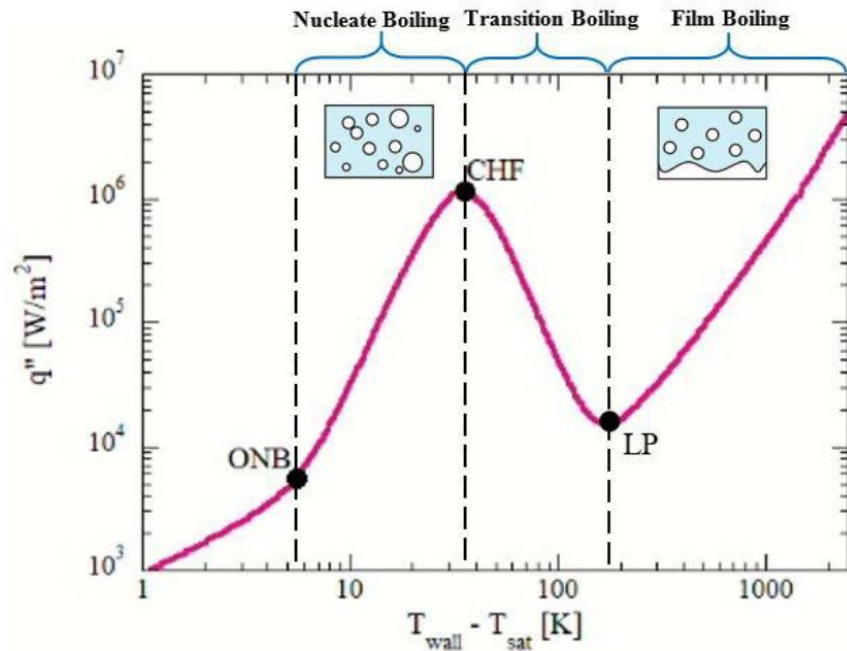
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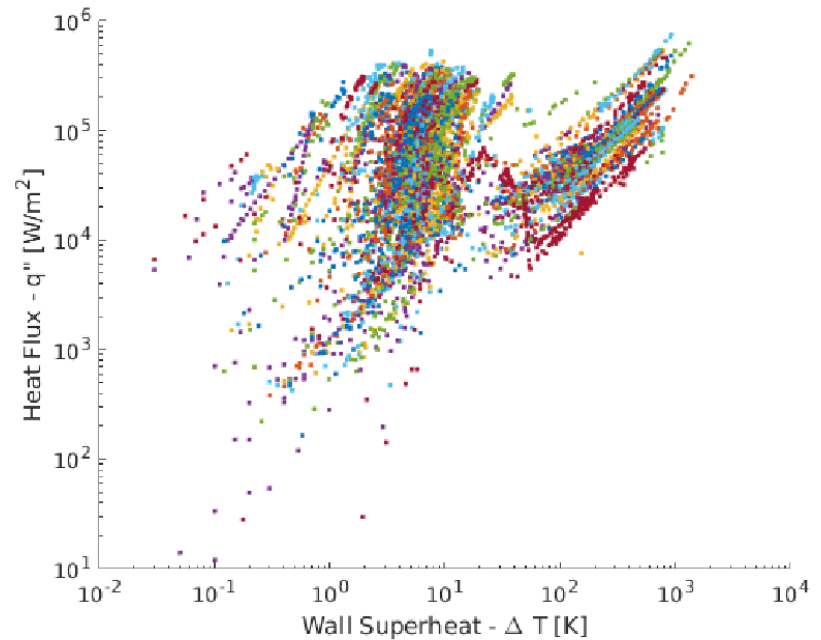
Background

- > Cryogenic liquids are used for propulsion and other space-based applications
- > Pool boiling is known to be a stable and efficient method of producing vapor and transferring large quantities of heat
- > In spite of the many studies of cryogenic boiling conducted to date, considerable experimental uncertainty in the results persists
- > Altering the texture of the surface has been seen to result in an increase in the rates of boiling and heat transfer

Pool Boiling – Constructing the Boiling Curve



“Typical” boiling curve

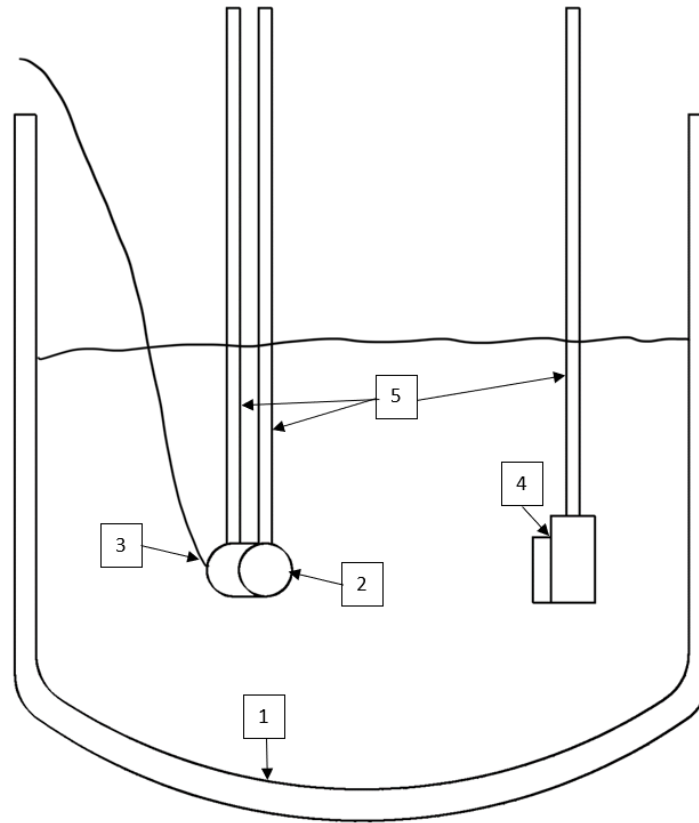


Actual boiling “curves”
(liquid nitrogen)

Current Work

- > Liquid nitrogen is employed given the large database of existing results
- > The configuration consists of an insulated vessel 13.5 inches (34.3 cm) in diameter fitted with horizontal test specimens
- > The test specimens consist of cylinders with smooth and grooved surfaces
- > Heating is provided by embedded electrical heaters
- > Copper-constantan thermocouples are used at inside and surface points to determine heat transfer
- > The grooved surface geometries employed here explore parametrically the impact on boiling and heat transfer

Test Apparatus



Experimental Configuration. (1), insulated container; (2), test surface (horizontal); (3), heater wire; (4), action camera; (5), mounting rods (light source not shown)

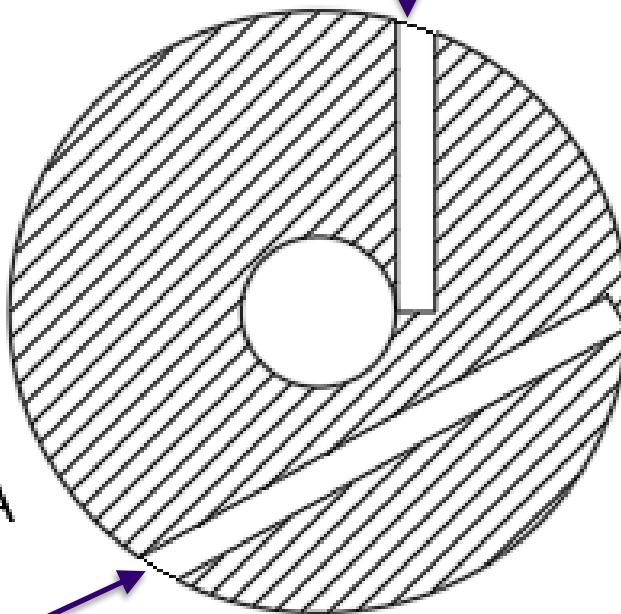
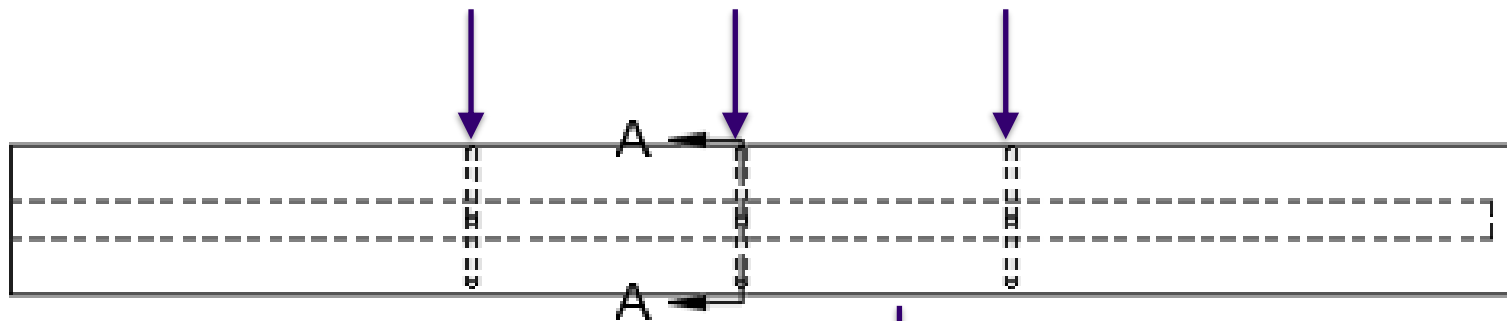
Test Apparatus



Test Conditions

- > Liquid nitrogen at atmospheric pressure ($T_{\infty} = 77 \text{ K}$)
- > Tests performed with aluminum cylinders 10.5" (26.7 cm) long and 1" (2.54 cm) in diameter
- > Smooth and grooved cylinders tested
- > Grooves were 2 mm wide, 1 mm deep with 1, 2, and 3 mm spacing
- > All tests conducted in the nucleate-boiling regime

Smooth Cylinder Geometry



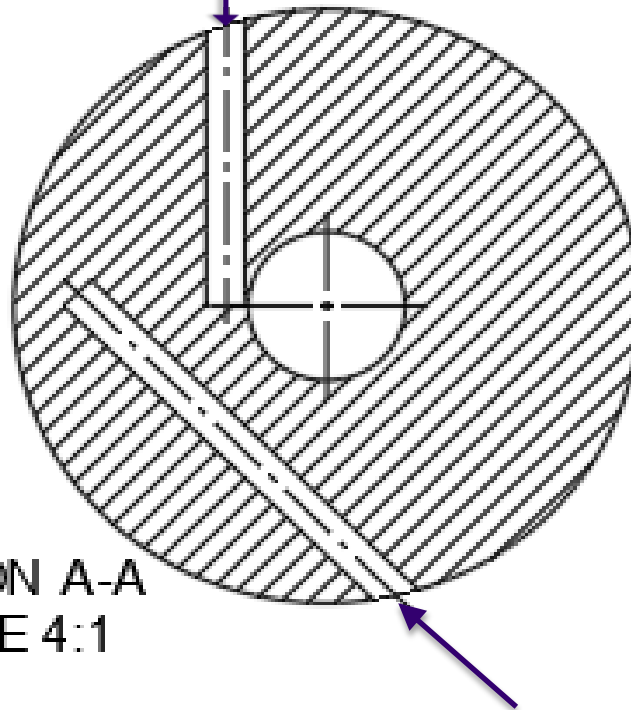
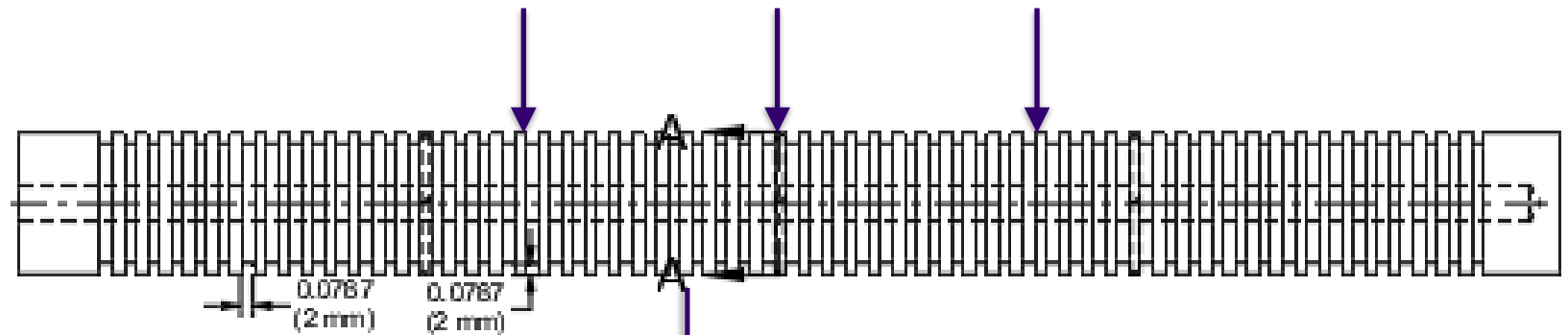
SECTION A-A
SCALE 4:1

Thermocouple well (6 total)

OD: 1"
ID: 0.25"
Length: 10.5"

Aluminum

Grooved Cylinder Geometry (2 mm groove spacing)



SECTION A-A
SCALE 4:1

OD: 1"
ID: 0.25"
Length: 10.5"

Aluminum

Surface: Regular circular
grooves, 2 mm thick
2 mm deep
2 mm spacing

Thermocouple well (6 total)

Test Cylinder Image



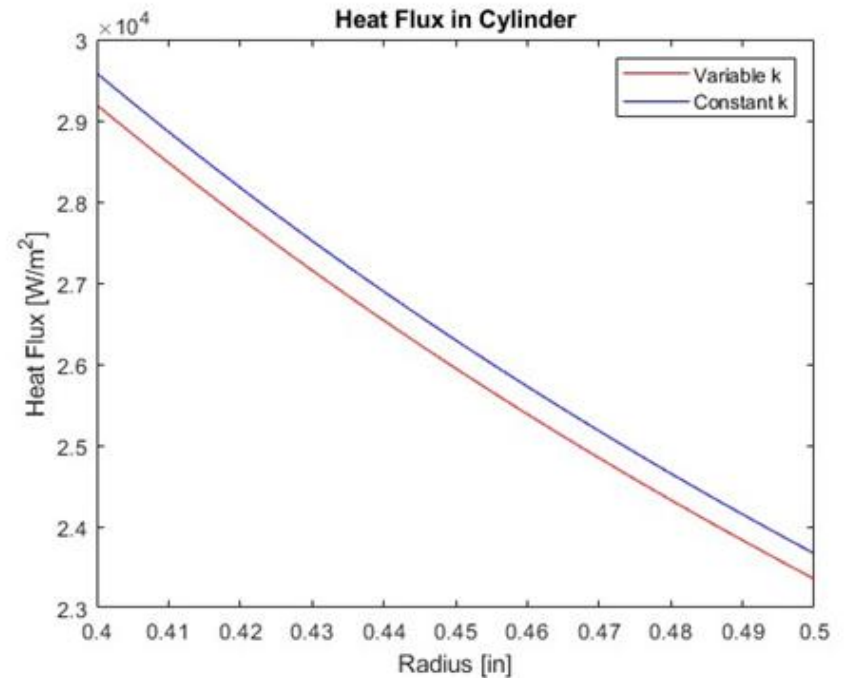
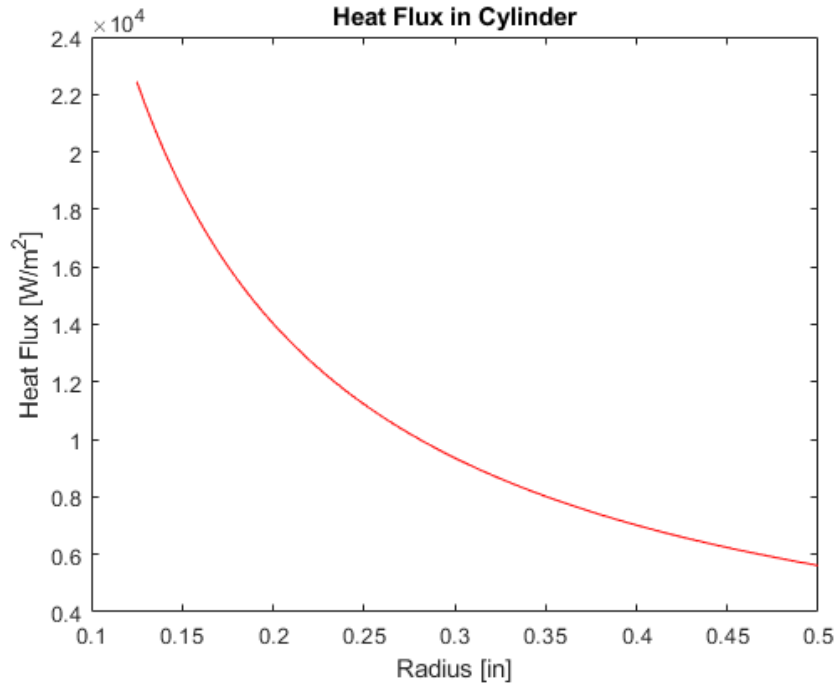
Grooved cylinder with 3 mm spacing

Heat Flux Calculations

- > 6 Copper-constantan thermocouples used at inside and surface points
- > Thermocouples used to calculate temperature profile in the cylinder
- > Variation in thermal conductivity with temperature considered to correct theoretical temperature profile to determine the surface heat flux

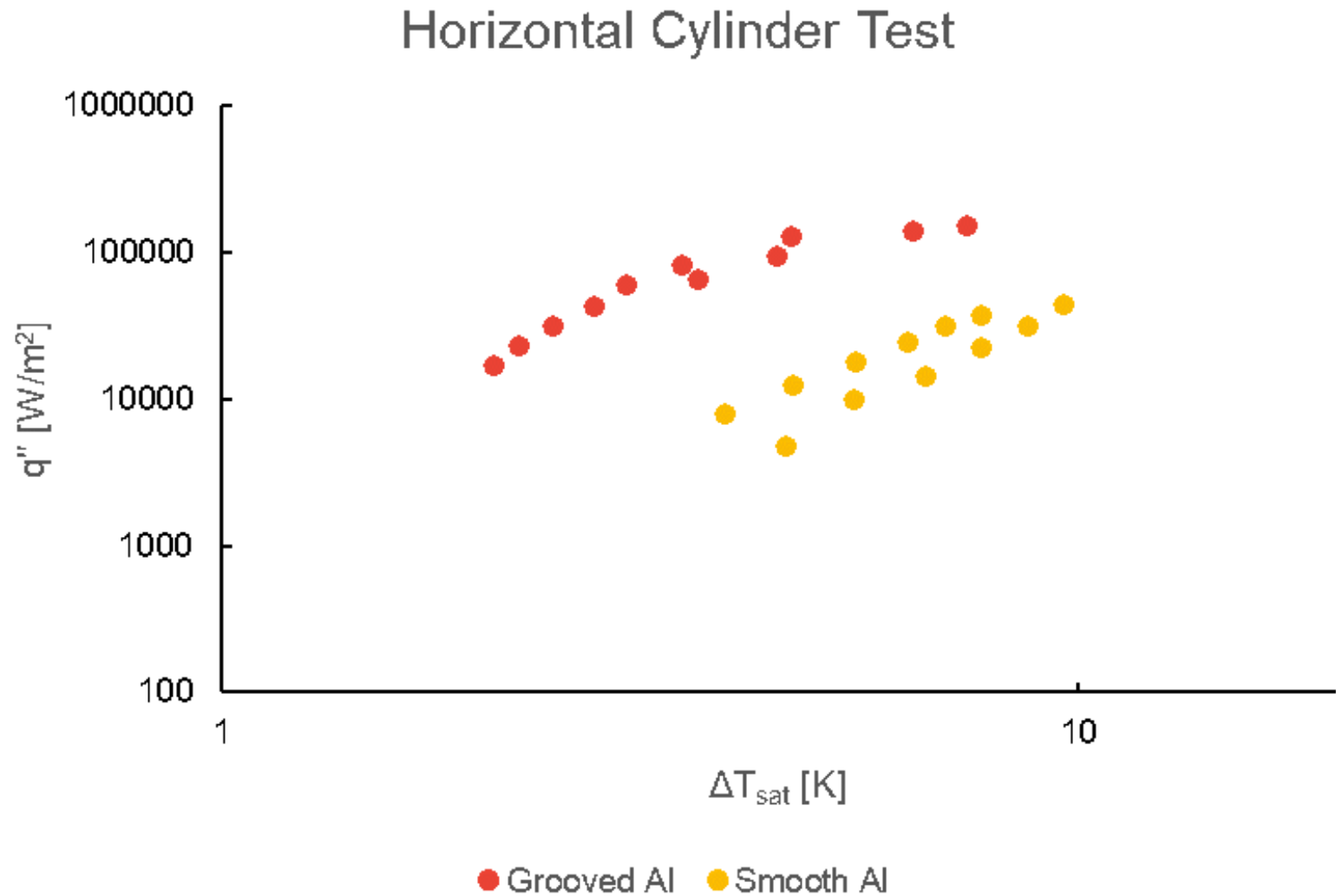
Heat Flux Determination

$$q = -k \frac{dT}{dr}$$

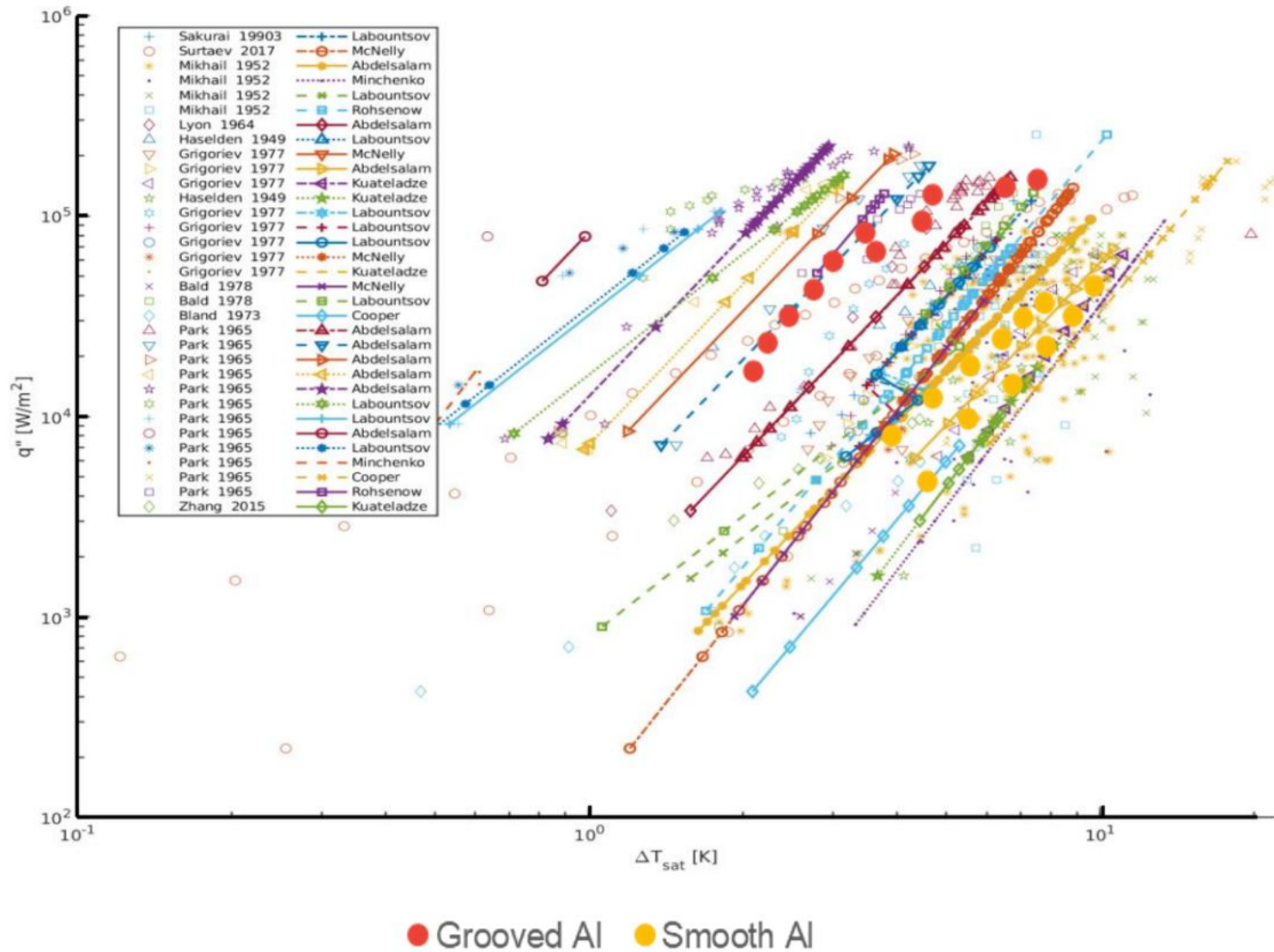


Overall computed heat flux decreased roughly 2% with variable thermal conductivity taken into account

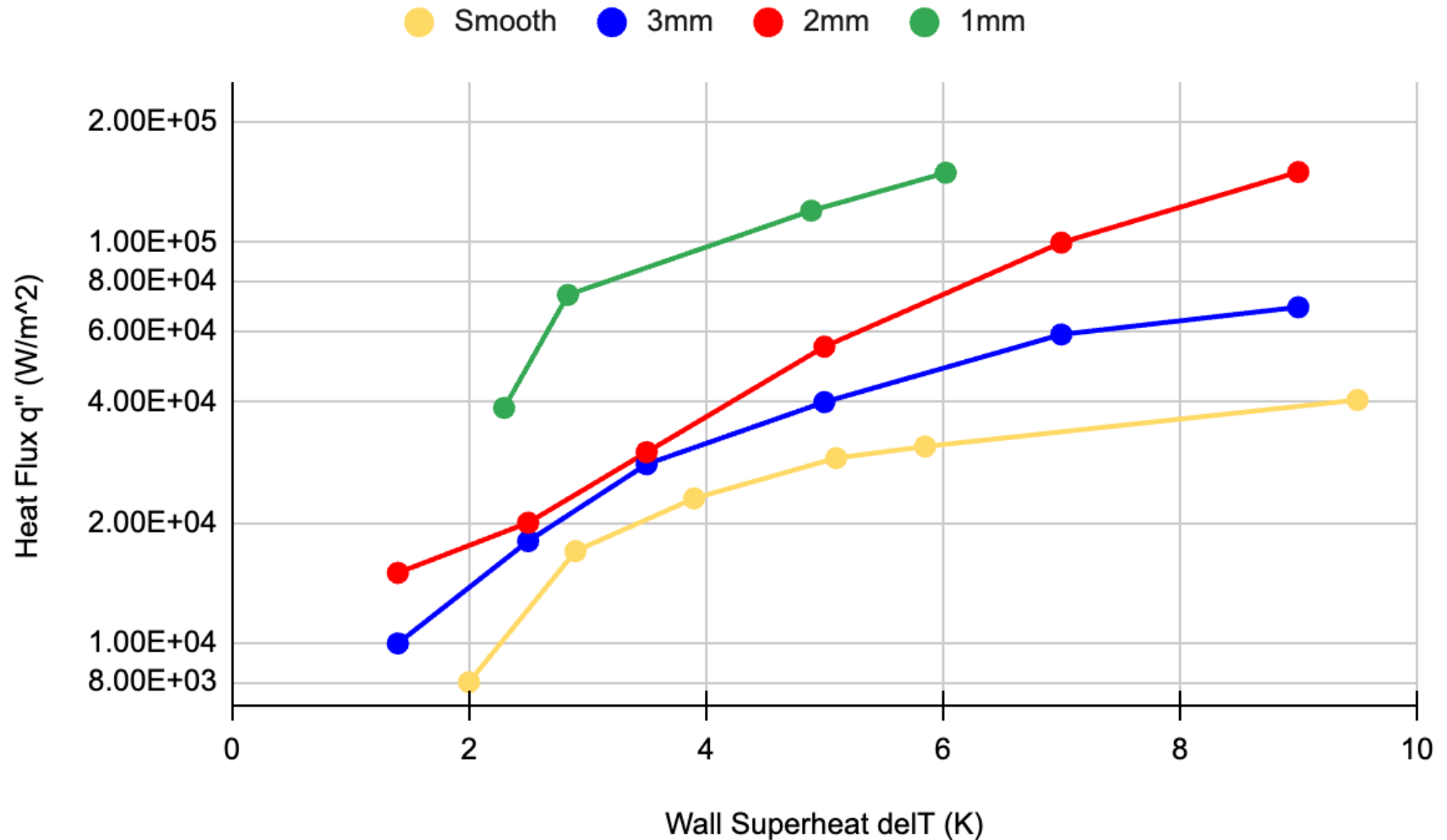
Results – 2 mm grooved cylinder



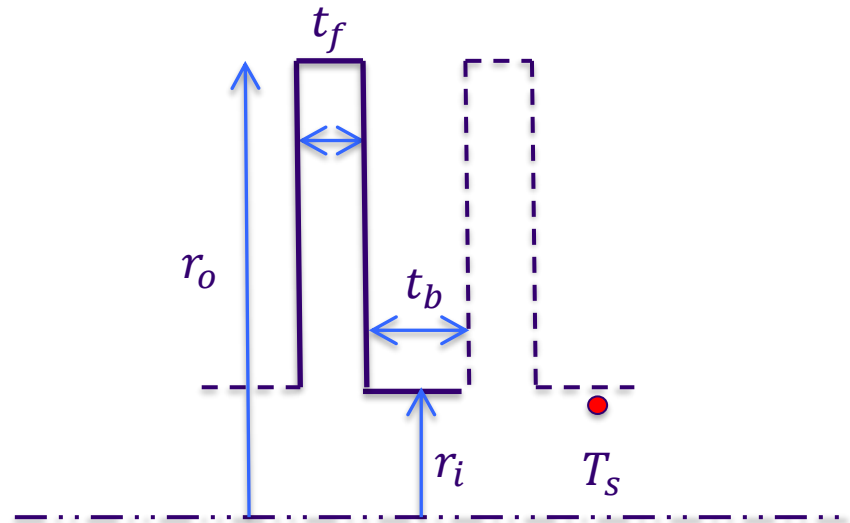
Horizontal Results Comparison



Wall Heat Flux for Different Surface Conditions



Heat Transfer Correction for Surface Area



Baseline (smooth surface)
heat transfer rate:

$$Q_{smooth} = h (T_s - T_\infty) A_{smooth}$$

$$A_{smooth} = 2\pi r_o (t_b + t_f)$$

Heat transfer for extended area,
assuming uniform temperature

$$Q_{sfc} = h_{sfc} (T_s - T_\infty) (A_{fins} + A_{base})$$

$$A_{base} = 2\pi r_i t_b$$

$$A_{fins} = 2\pi (r_o^2 - r_i^2) + 2\pi r_o t_f$$

At given test condition:

$$\frac{h_{sfc}}{h} = \frac{r_o (t_b + t_f)}{(r_o^2 - r_i^2) + r_o t_f + r_i t_b}$$

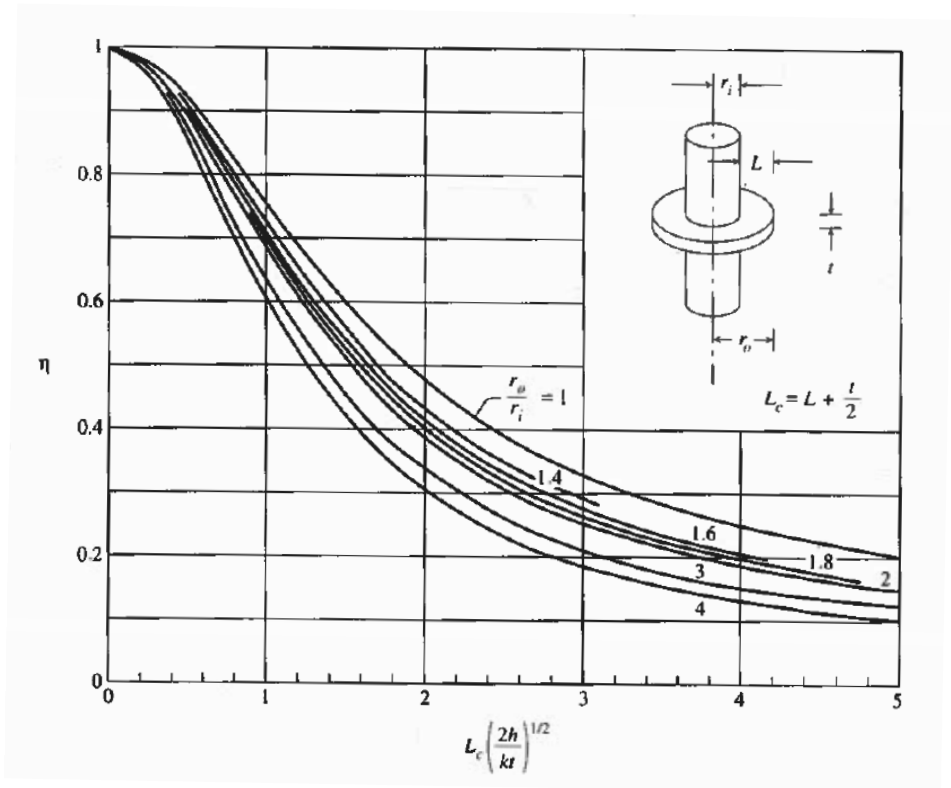
Heat Transfer Correction for Fin Effectiveness

Fin Efficiency

$$\eta = \frac{Q_{fin}}{h_{sfc} A_{fin} (T_s - T_\infty)}$$

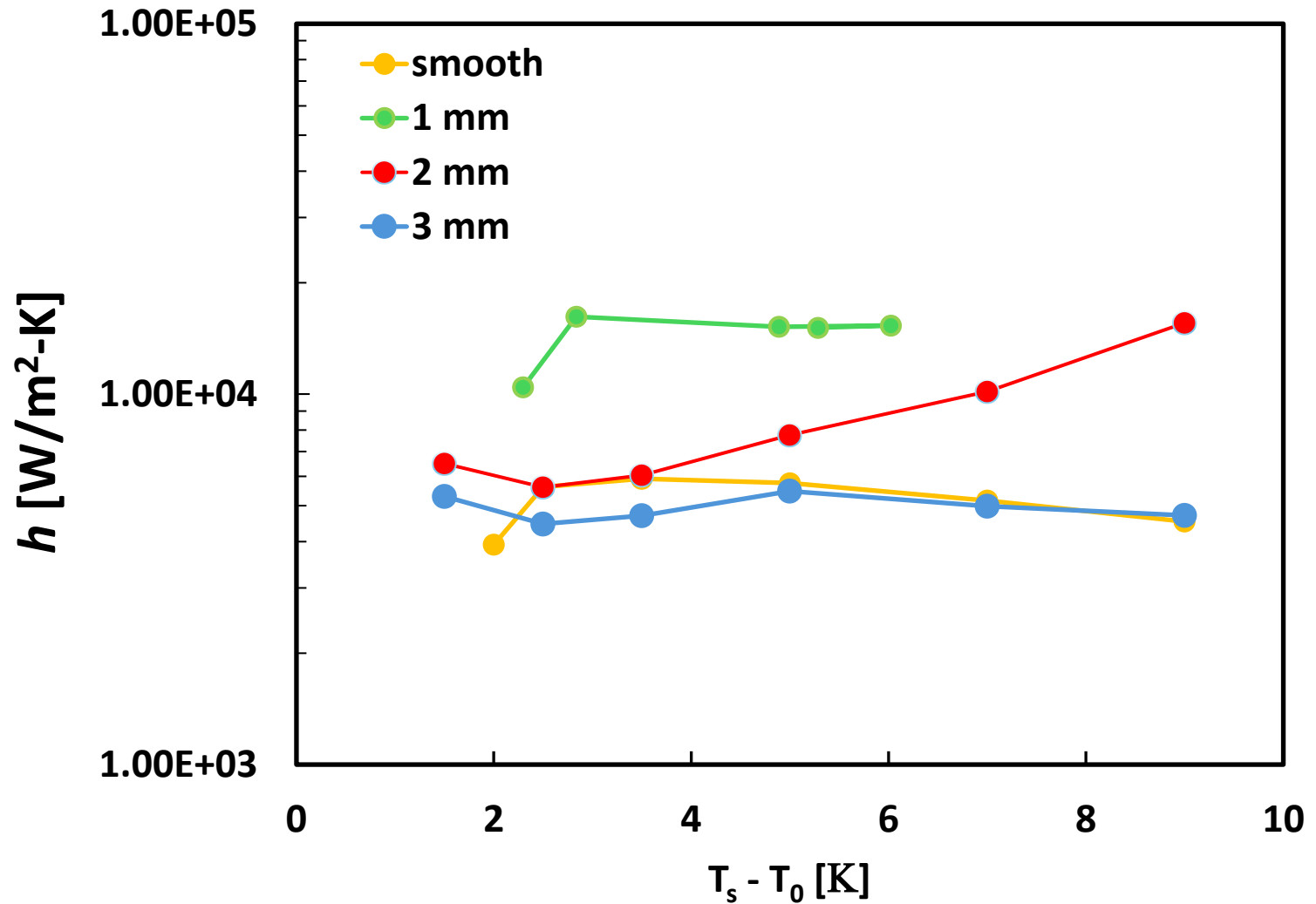
$$\begin{aligned} Q_{corr} &= h_{sfc} (T_s - T_\infty) [\eta A_{fin} + A_{base}] \\ &= h_{corr} (T_s - T_\infty) A_{smooth} \end{aligned}$$

$$\Rightarrow \frac{r_i(t_b + t_f)}{\eta(r_o^2 - r_i^2 + r_o t_f) + r_i t_b} = \frac{h_{corr}}{h_{sfc}}$$



Gardner, K. A., "Efficiency of extended surfaces," *Transactions of the American Society of Mechanical Engineers*, Vol. 67, No. 8, 1945, pp. 621–628.

Corrected Heat Transfer Coefficient

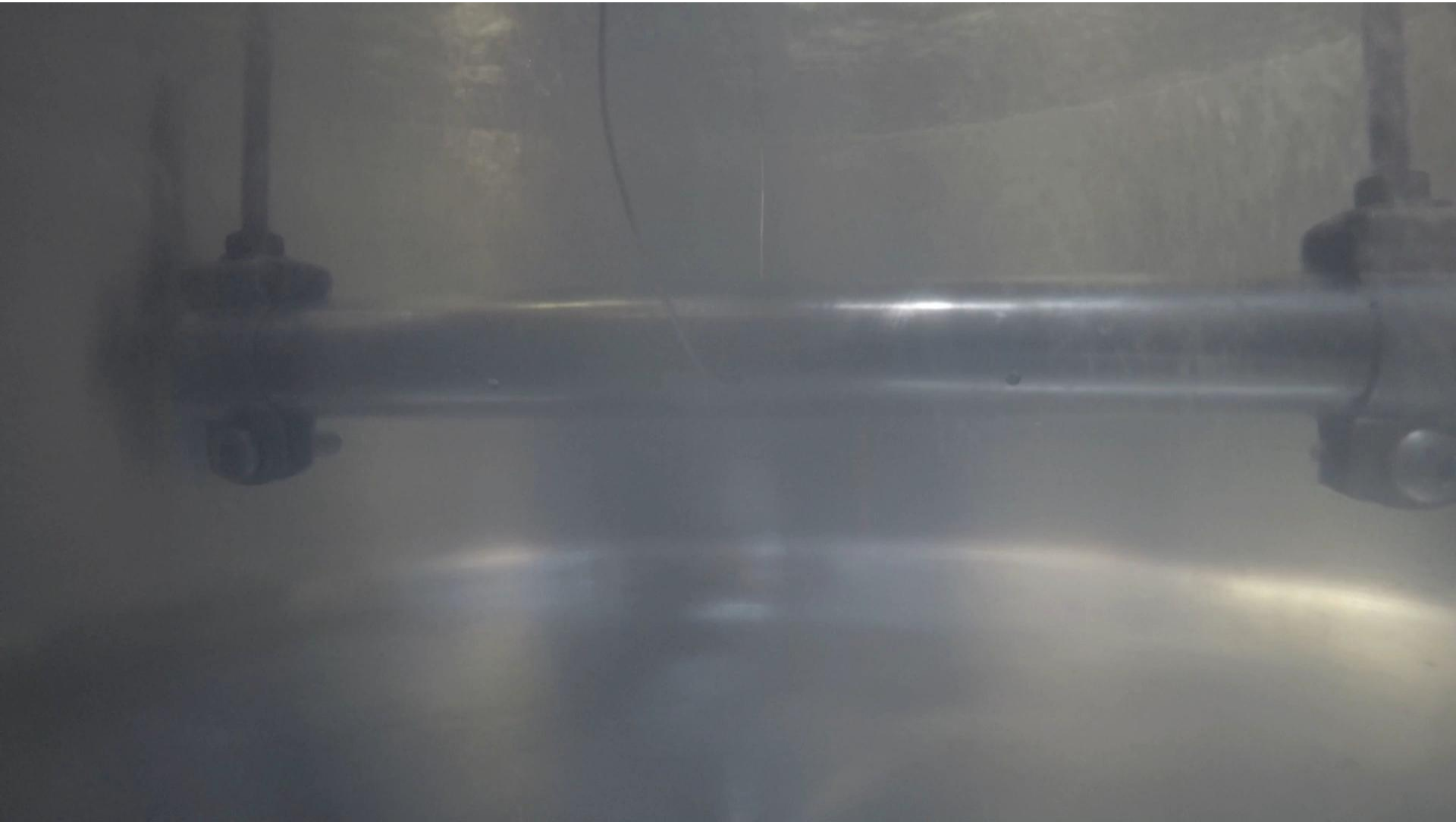


Tabular Summary of Results

Heat transfer enhancement factors
due to the addition of grooves

Horizontal Cylinder	Heat flux	Heat transfer coefficient
3-mm groove spacing	1.8-1.9x	0.8-1.0x
2-mm groove spacing	2.1-3.3x	1.0-3.4x
1-mm groove spacing	3.3-5.0x	1.7-2.8x

Video of Boiling



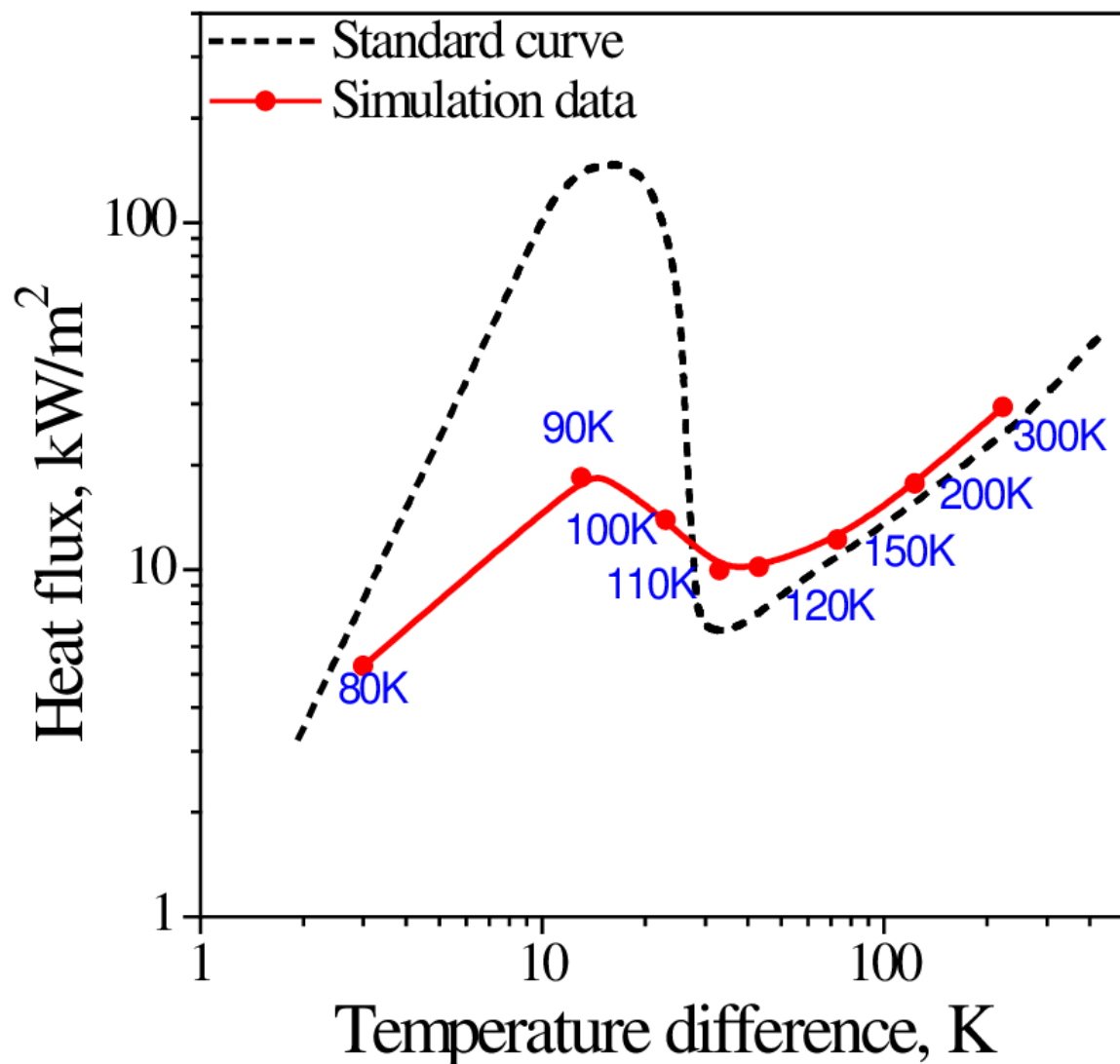
Summary and Conclusions

- > Experiments were conducted to evaluate the effects of surface modifications on the nucleate pool boiling of liquid nitrogen.
- > Temperature probes, combined with the variable thermal conductivity of the solid, were used to calculate the heat flux at the surface.
- > Grooved surfaces generally resulted in an increase in the heat flux, by a factor of up to 5.
- > The degree of heat flux enhancement increased with decreasing groove spacing .
- > The estimated heat transfer coefficients were higher for the narrower groove spacing.
- > The 1 mm groove spacing, even though smaller than the estimated break-off vapor bubble diameter, still results in an increase in heat transfer over the smooth cylinder.

Questions/comments?

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Example TC Output Horizontal Cylinder

